NIFTY-50 Sectoral Performance & Risk Intelligence

NST DVA Capstone 2 Team | Finance / Stock Market Analytics

Slide 2. Problem Statement

- Raw market data is not decision-ready on its own
- Investors need return, risk, liquidity, and conviction in one framework
- Project goal: identify which stocks and sectors offer the strongest balance between opportunity and risk

Slide 3. Dataset Overview

- Primary source: Kaggle NIFTY-50 Stock Market Data (2000-2021)
- Cleaned coverage: 235,192 rows
- Stable stock keys: 49
- Historical row-level symbols: 65
- Sector count: 13
- Date range: 2000-01-03 to 2021-04-30

Slide 4. ETL And Cleaning

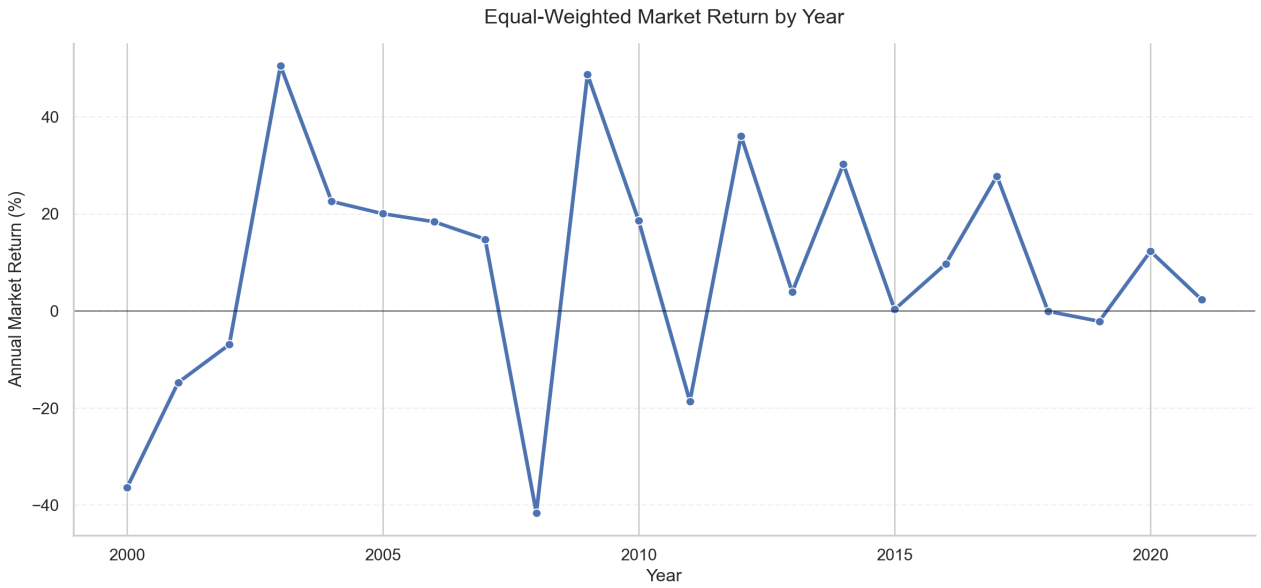
- automatic file detection and metadata merge
- column standardization to snake_case
- turnover rescaling by 100000
- quality flags for missing prices, missing volume fields, invalid OHLC, and outlier returns
- important remaining gaps: 114,848 missing trades rows and 16,077 missing delivery fields

Slide 5. KPI Framework

- return: cumulative return, annualized return, average daily return
- risk: volatility, max drawdown, risk-adjusted return
- liquidity: average volume, average turnover, liquidity score
- conviction: average delivery percent, close above VWAP ratio, investor confidence score
- screening metric: opportunity score

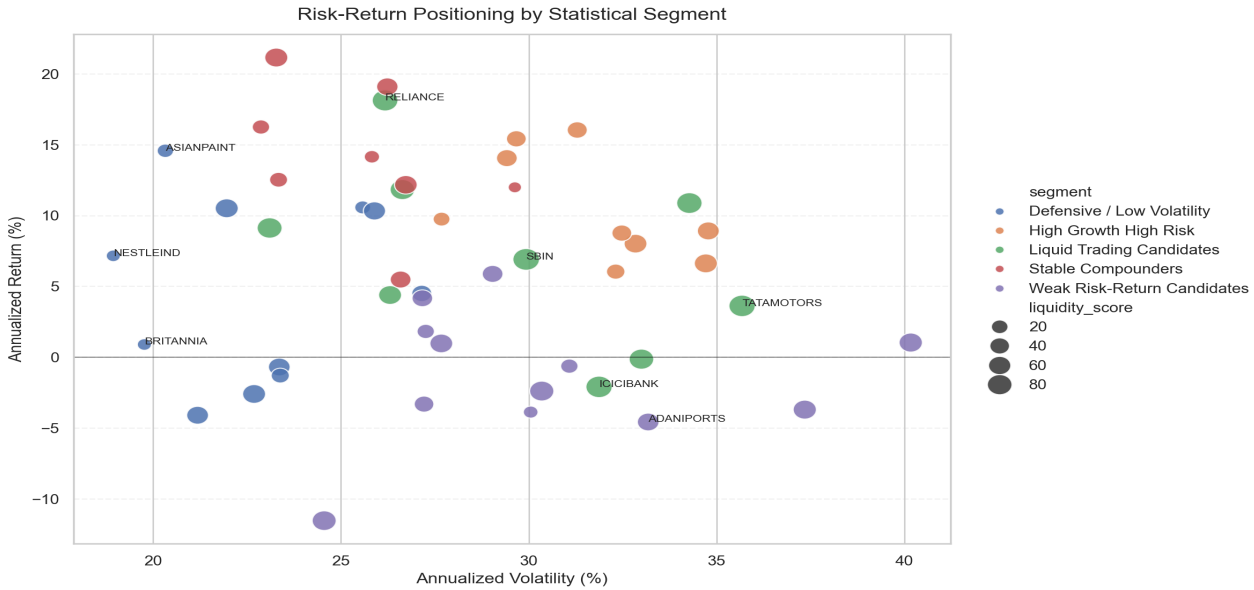
Slide 6. EDA Insights

- best yearly expansion: 2003 at 50.54%
- worst yearly drawdown: 2008 at -41.59%
- COVID crash: -17.25%
- COVID recovery: 37.42%
- cement, IT, and pharma emerge as strong sector stories



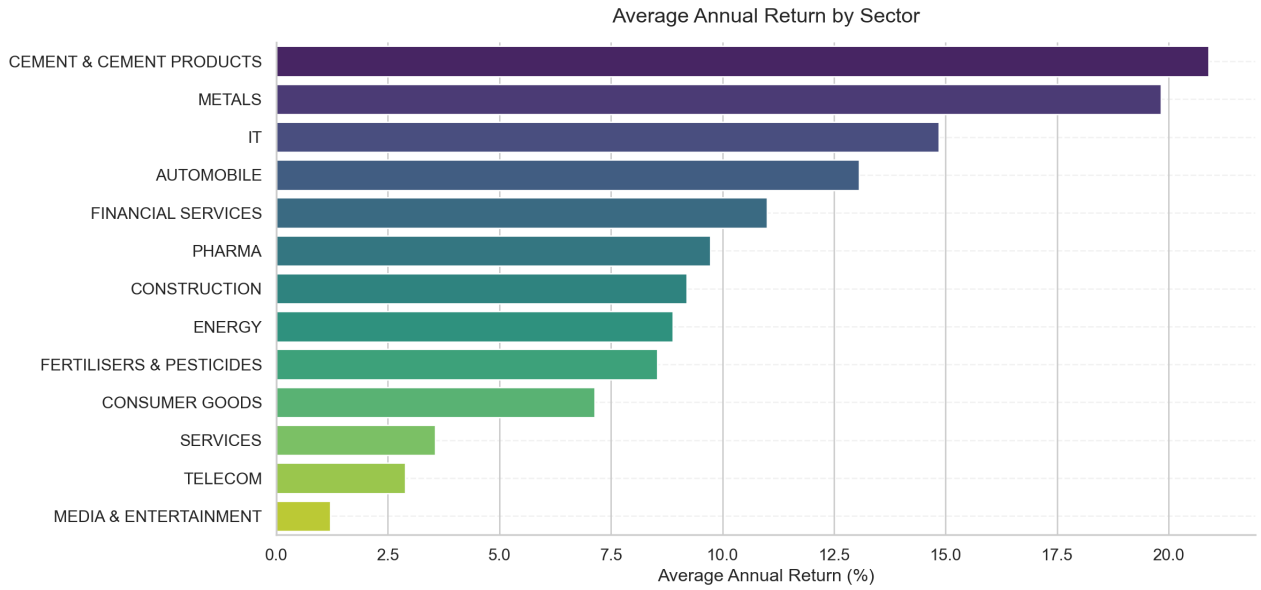
Slide 7. Statistical Analysis

- weak but statistically significant daily-return links with volume and turnover
- delivery intensity did not significantly change stock-level return averages
- sector-return differences were statistically detectable: p = 0.0379
- risk layer adds drawdown, downside volatility, VaR, and CVaR
- segmentation produces five explainable action groups



Slide 8. Tableau Dashboard Walkthrough

- executive KPI cards
- market trend section
- sector comparison section
- stock opportunity scatter and ranking
- COVID crash vs recovery comparison
- recommendation action board



Slide 9. Key Recommendations

- use Buy names as the core shortlist
- monitor Watch names for defensive positioning
- use Trade Candidate names tactically, not blindly
- keep Avoid names below the fold in default ranking
- lead sector discussion with cement, IT, pharma, automobile, and financial services

Slide 10. Impact, Limitations, And Future Scope

- decision-support impact: first-pass stock shortlist reduced from 49 names to 8 Buy names
- limitations: historical sample, missing delivery/trades fields, non-causal statistics
- future scope: benchmark comparison, rolling-window tests, valuation overlays, published Tableau story

Slide 11. Contribution Matrix

- Atharv: setup, sourcing, problem framing
- Devaansh: ETL and cleaned dataset
- Aryan: EDA, KPIs, chart outputs
- Abhey: statistics, risk, segmentation
- Birajit: final Tableau pack, report, presentation, QA, submission readiness

Slide 12. Viva Preparation

- be ready to justify dataset choice
- explain why turnover was rescaled
- explain why rule-based segmentation was preferred over forced clustering
- explain why significant p-values can still be economically weak
- explain why the dashboard is designed for decision support instead of decoration